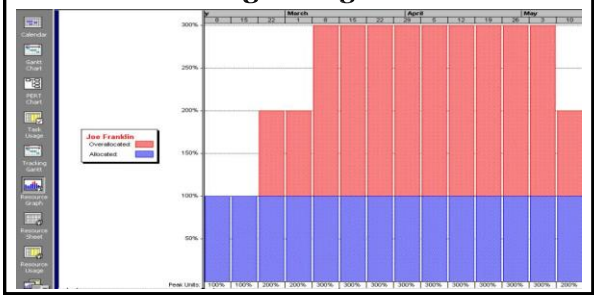


Topic # 8
Project Resource Allocation, Monitoring & Project Control
Chapter 8 – Schwalbe

- ***Resource loading***
  - Refers to the amount of individual resources an existing schedule requires during specific time periods
- Helps project managers develop a general understanding of the demands a project will make on the organization's resources and individual people's schedules
- Project Managers use ***histograms*** to depict period by period variations in resource loading.
- ***Over allocation***
  - More resources than are available are assigned to perform work at a given time

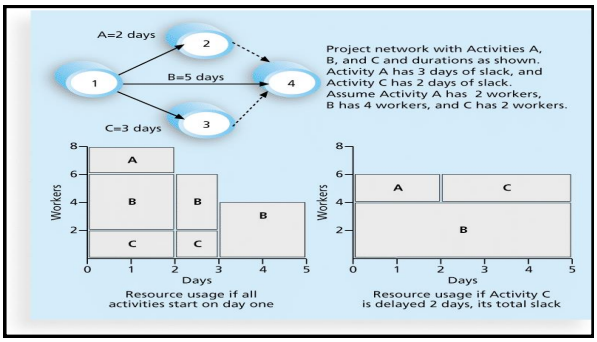
Resource Loading Histogram





Resource Leveling

- A technique for resolving resource conflicts by delaying tasks
- The main purpose of resource leveling is to create a smoother distribution of resource usage and reduce over allocation.
- Project Managers examine the network diagram for areas of slack and to identify resource conflicts.



Resource Leveling: Advantages

- When resources are used on a more constant basis, they require less management.
- It may enable project managers to use a just-in-time inventory type of policy for using subcontractors or other expensive resources.
- It results in fewer problems for project personnel and accounting department
- It often improves morale

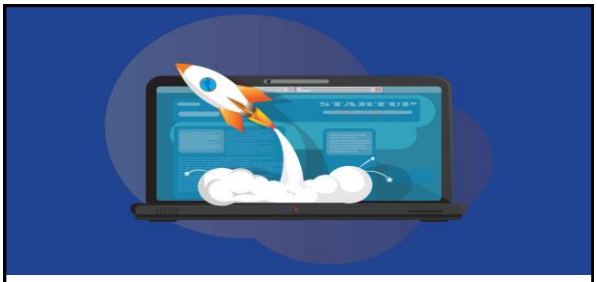
- Project Management software can automatically level resources.
- The project manager must be careful in using the results without making adjustments.
- Different algorithms can produce different leveling results.



Shortening Schedule (Using Critical Path)

Techniques for shortening schedules

- Shortening durations of critical activities/tasks by adding more resources or changing their scope
- **Crashing** activities by obtaining the greatest amount of schedule compression for the least incremental cost
  - making cost & time tradeoff.
- **Fast tracking** involves doing activities in parallel that would normally be done in sequence



Project Crashing

- Projects will sometimes have deadlines that are impossible to meet using normal procedures
- By using exceptional methods it may be possible to finish the project in less time than normally required
- However, this usually increases the cost of the project
- Reducing a project's completion time is called *crashing*
  - *Normal Time*
  - *Normal Cost*
  - *Crash Time*
  - *Crash Cost*

Project Crashing – FOUR STEPS

1.Find the normal critical path and identify the critical activities

1.Compute the crash cost per week (or other time period) for all activities in the network using the formula

$$\text{Crash cost/Time period} = \frac{\text{Crash cost} - \text{Normal cost}}{\text{Normal time} - \text{Crash time}}$$

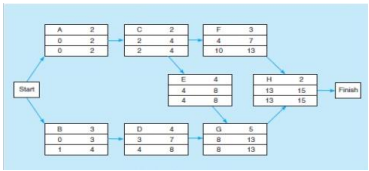
3.Select the activity on the critical path with the smallest crash cost per time period and crash this activity to the maximum extent possible or to the point at which your desired deadline has been reached

3.Check to be sure that the critical path you were crashing is still critical. If the critical path is still the longest path through the network, return to step 3. If not, find the new critical path and return to step 2.

Example

- General Foundry has been given 14 weeks instead of 16 weeks to install the new equipment
- The critical path for the project is 15 weeks
- What options do they have?
- The normal and crash times and costs are shown in Table
- Crash costs are assumed to be linear and Figure shows the crash cost for activity B
- Crashing activity A will shorten the completion time to 14 but it creates a second critical path B,D,G,H.
- Any further crashing must be done to both critical paths

Example

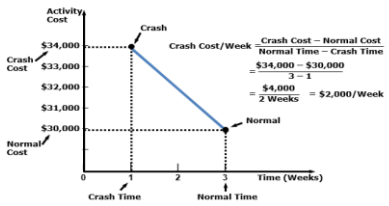


- Normal and crash data

ACTIVITY	TIME (WEEKS) NORMAL	TIME (WEEKS) CRASH	COST ($) NORMAL	COST ($) CRASH	CRASH COST PER WEEK ($)	CRITICAL PATH?
A	2	1	22,000	23,000	1,000	Yes
B	3	1	30,000	34,000	2,000	No
C	2	1	26,000	27,000	1,000	Yes
D	4	3	48,000	49,000	1,000	No
E	4	2	56,000	58,000	1,000	Yes
F	3	2	30,000	30,500	500	No
G	5	2	80,000	86,000	2,000	Yes
H	2	1	16,000	19,000	3,000	Yes

Example

- Crash and normal times and costs for activity *B*





Scope Creep

- SCOPE CREEP:
  - Scope creep is also known as requirement creep, which refers to the uncontrolled changes in the project's or product's scope.
  - Scope creep happens in the project for following reasons:
    - Due to interference from the client.
    - Due to an incomplete scope statement.
    - Due to a poor change control system.
    - Due to miss-communication among the team members.
    - Due to reasons external to organizations; e.g. market conditions, regulatory requirements, or technological advancements.
  - Avoiding Scope Creep & Issues in Controlling (Discussed in Detail in Project Scope Management)



Gold Plating

- Gold plating means intentionally adding extra features or functions to the products which were not included in the scope statement.
- Usually gold plating is performed by either the project team or the project manager with no additional cost to the client.
- Following are a few causes of gold plating:
  - Team member may add extra functions to prove his abilities to the project manager.
  - Project manager may add extra functions to earn credit from the client or the top management.
  - Sometimes it is performed to divert the attention of the client from the defects in the product.
- Gold plating increases the input cost, increases the risk, and the expectation of the customer is elevated.
- Gold Plating can be avoided:
  - Team member shouldn't be allowed to add any extra function or features to the product without approval.
  - As a project manager one should also avoid it.
  - By establishing proper communication lines within the project team.

END OF TOPIC 8

- COMING UP!!!!!!
- Risk Management
- HR Management
- Midterm II
- Agile Project Management